AGRIKRANTI 1928
Aditya Kumar, M.Sc. (Ag.) Soil Science - Gold Medalist (DDUGU) · Semester Paper Series

B.Sc. (Hons.) Agriculture (First Semester) Examination, 2021-22

MAT-101: Elementary Mathematics · Credit 2 (1+1)

Previous Year Question Paper (Paper XI, Code 639(A)) - re-typed for practice

Time: 2 Hours **Maximum Marks: 70**

Note: (i) Attempt any **five** questions from Section A and any **three** questions from Section B. (ii) Answer each question of Section A within 50 words. खंड A से कोई पाँच तथा खंड B से कोई तीन प्रश्न कीजिए।

Section A (Attempt any five) — 5 × 5 = 25

1. Find the coordinates of the point which divides internally the join of the points (8, 9) and (-7, 4) in the ratio 2 : 3. **5**

2. Find the area of the quadrilateral whose vertices are A (-4, 5), B (0, 7), C (5, -5) and D (-4, -2). **5**

3. Show that the straight lines $27x - 18y + 25 = 0$ and $2x + 3y + 7 = 0$ are perpendicular to each other. **5**

4. Find the equations of tangent and normal to the circle $x^2 + y^2 = 25$ at (-3, 4). **5**

5. Prove that: $\lim_{x \to 0} \frac{\log (1 + x)}{x} = 1$ **5**

6. Prove that $f(x) = |x|$ is continuous at $x = 0$. **5**

7. Differentiate $e^{\tan x}$ with respect to x. **5**

8. Find the maximum and minimum value of the polynomial $y = 3x^4 - 2x^3 - 6x^2 + 6x + 1$. **5**

Section B (Attempt any three) — 3 × 15 = 45

1. Evaluate: (i) $\int x \sqrt{(1 + x)}\, dx$ (ii) $\int_0^1 (1 + x^2)^3\, dx$ **15**

2. Using integration, find the areas of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and the circle $x^2 + y^2 = 9$. **15**

3. Using the properties of determinants, prove that: $\begin{vmatrix} x^2 & y^2 & z^2 \\ yz & zx & xy \\ x & y & z \end{vmatrix} = (x - y)(y - z)(z - x)(xy + yz + zx)$ **15**

4. If $A = \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix}$ and $B = \begin{vmatrix} 0 & 1 \\ 2 & 4 \end{vmatrix}$, then verify that $(AB)^{-1} = B^{-1}A^{-1}$. **15**

5. (i) Find the equation of straight line which passes through the point (3, 4) and the sum of whose intercepts on the axes is 14. (ii) Find the equation of the circle of radius 5 units, whose centre lies on the x-axis and which passes through the point (2, 3). **15**